<u>605.201 Mini-Project 3:</u>

Please do the following to complete this assignment.

Purpose:

The purpose of this project is to provide non-trivial practice in the use of Java object-oriented GUI programming features to implement an object-oriented GUI design and have a bit of fun doing it.

Resources Needed:

You will need a computer system with Java 7 or greater SE edition run-time and Java Development Kit (JDK). You may optionally use a Java IDE for example NetBeans, Eclipse, etc. However application builders are not allowed.

Submitted Files:

Source files:

Each public class *must* be contained in a separate Java source file.

The format of the Java source must meet the general Java coding style guidelines discussed so far during the course. Pay special attention to naming guidelines, use of appropriate variable names and types, variable scope (public, private, protected, etc.), indentation, and comments. Please use course office hours or contact the instructor directly if there are any coding style questions.

Submit file:

The submit file is to be a Zip file containing your Java sources, and your Javadocs directory/folder. Any appropriate file name for this Zip file is acceptable.

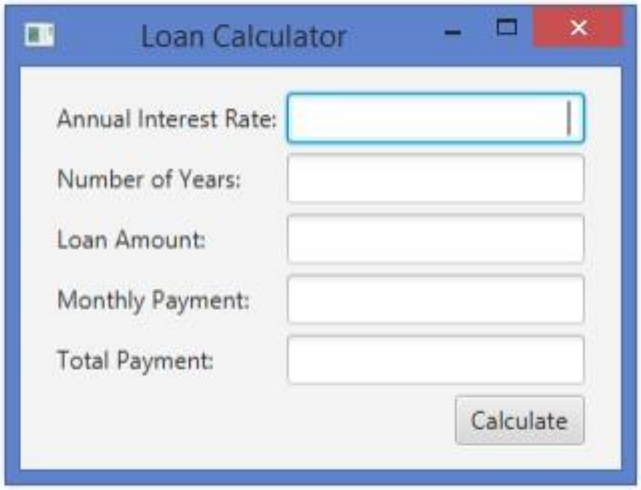
If you know how to create a standard Java JAR file, this is also acceptable for your source code. However, make sure you include the source code in your JAR file.

Collaboration:

It is encouraged to discuss technical or small design parts of this project with your fellow students. However, the resulting design and implementation must be your own. For example, it is acceptable to discuss different ways of maintaining the system state but not detailed design or implementation information on processing the loan calculation. When in doubt, ask during office hours or contact your instructor.

Program Specification:

This project involves implementing a Java program that performs a calculation of payments associated with loans. The program shall allow a user to enter the following data: annual interest rate, the term of the loan (i.e., number of years), and the loan amount. A GUI like the one below must be used.



When the user presses the calculate button, the monthly payment and total amount of payments shall be displayed.

The formula for computing the monthly payment (*P*) is:

$$\frac{i*A}{1-(1+i)-n}$$

where *i* = periodic interest rate, *A* = amount of loan, *n* = number of compounding periods (12).

For example, if the annual interest rate is 5 percent, the term of the loan is 2 years, and the loan amount is $2,000, and interest is compounded monthly:

i = .05/12 = .00417   n = 2 * 12 = 24   P = $87.74   Total Payment = 24 * $87.74

Assignment Rubric:

Part	70%	80%	90%	100%	% grade
Functionality	Majority of required function parts work as indicted in the assignment text. One major or 3 minor defects. All major functionality at least partially working (example change provided but not correct).	Most required function parts work as indicted in the assignment text above and submitted documentation. One major or 3 minor defects. All major functionality at least partially working (e.g., payment calculated, but not correct).	Nearly all required function parts work as indicted in the assignment text above and submitted documentation. One to two minor defects.	All required function parts work as indicted in the assignment text above and submitted documentation.	65%
Code	Majority of the code conforms to coding standards as explained and demonstrated so far in the course (ex. method design, naming, formatting, etc.). Five to six minor coding standard violations. Some useful comments. Code compiles with multiple warnings or fails to compile with difficult to diagnose error.	Most of the code conforms to coding standards as explained and demonstrated so far in the course (ex. method design, naming, formatting, etc.). Three to four minor coding standard violations. Mostly useful comments. Code compiles with one to two warnings.	Almost all code conforms to coding standards as explained and demonstrated so far in the course (ex. method design, naming, formatting, etc.). One to two minor coding standard violations. Appropriate level of useful comments. Code compiles with no errors or warnings.	All code conforms to coding standards as explained and demonstrated so far in the course (ex. method design, naming, formatting, etc.). Appropriate level of useful comments. Code compiles with no errors or warnings.	25%
Submit package	More than one file submitted in incorrect format. Files not enclosed in the specified compressed file.	All but one file submitted in correct format. Files not enclosed in the specified compressed file.	All file submitted in correct format but not in the specified compressed file.	All file submitted in correct file formats and compressed as specified.	10%